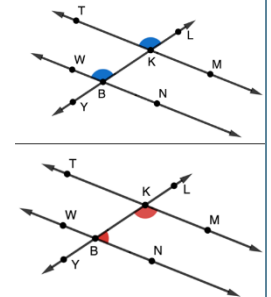


3.3 Congruent Angles in Parallel Lines

Lesson 3 Introduction

 Benchmark Focus	MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.		 Learning Target	I can use postulates and theorems to find unknown angle measures when parallel lines are intersected by a transversal.
 Benchmark Coherence	Building On Working Toward MA.8.GR.1.4: Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.		Working Toward N/A	
 Essential Questions	- What are the names of the angle pairs formed by parallel lines intersected by a transversal? - What is the relationship between angle pairs formed by two parallel lines intersected by a transversal?			
 Lesson Narrative	This lesson begins with students using patty paper to determine the relationships between angles formed by two parallel lines and a transversal. Students are introduced to alternate interior and alternate exterior angles. They identify angle pairs and complete a formal proof of the Alternate Exterior Angle Theorem. Practice centers around determining the measures of other angles when given the measure of one angle formed by parallel lines and transversals.			
 Component Summary	3.3.1 Warm-Up (~5 min.)	Career profile video using math to make guitars (<i>SE p. 139</i>)	3.3.3 Try It (10 min.)	Determine angle measures when parallel lines are intersected by a transversal (<i>SE p. 142</i>)
	3.3.2 Guided Instruction (15-20 min.)	Explore alternate interior and alternate exterior angle relationships (<i>SE p. 139</i>)	3.3.4 Your Turn! (5 min.)	Practice determining angle measures when parallel lines are intersected by a transversal (<i>SE p. 143</i>)
	3.3.5 Wrap-Up (~5 min.)	Ticket Out the Door identifying angle relationships and using to find angle measure (<i>SE p. 143</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Alternate Exterior Angles Theorem - Alternate Interior Angles Theorem <u>Additional Terms:</u> N/A		- Patty Paper (Optional) Highlighters	3.3.1 Warm-Up includes the students watching a career profile video. - Consider providing a word bank for students to choose from for the proof in 3.3.2 Guided Instruction.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse the name of alternate interior and alternate exterior angles, or the relationship between the angle pairs. - Students may assume the same angle relationships exist between angles formed by two different transversals.	Congruence of the angle pairs only exist when the lines intersected by the transversal are parallel. Consider providing examples of non-parallel lines intersected by a transversal to show that alternate interior angles, alternate exterior angles, and corresponding angles do not have congruence in this case.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 3.3.3 Try It question 1 offers an opportunity for students to share what they notice and wonder about a given figure before completing a table. Allowing students to make sense of the image prior to answering the questions provides an entry-point into the question.	MA.K12.MTR.1.1: Participate in Effortful Learning Question 2 of 3.3.3 Try It involves parallel lines with two intersecting transversals. Offering a challenging task fosters perseverance as students analyze and make sense of the problem. Students should be encouraged by recognizing their efforts when presented a challenge. Use this as an opportunity to develop students' ability to analyze and problem-solve based on the relationships they have learned.
 Differentiate Instruction	The use of patty paper in the 3.3.2 Guided Instruction creates visual and kinesthetic entry-points as students make sense of the angle relationships formed by two parallel lines intersected by a transversal. Some students struggle with accurately tracing images and may benefit from already having two angles drawn to match the two angle measures shown in the figure.	



Lesson Notes:

3.3.1 Warm-Up: Using Math in Making Guitars

Component Overview: Students watch a career video of how mathematics is used in building guitars.



3.3.1 Warm-Up: Using Math in Making Guitars

Patrick Evans is a production manager at Maton Guitars in Australia. Country music star Keith Urban uses Maton guitars. The precision necessary in making each guitar requires mathematics, or “maths,” as Australians call it.



1. What was one thing you found interesting about the process of making a guitar?

Answers will vary.



Consider having the career profile video “Maths: Make your career count - Guitar Maker” (4:17 minutes) preloaded to avoid potential technology challenges.



What other connections can you make between music and mathematics?

3.3.2 Guided Instruction: Angle Relationships

Component Overview: Students investigate the relationship between angles formed by two parallel lines intersected by a transversal before being introduced to the Alternate Exterior Angles Theorem and the Alternate Interior Angles Theorem. After identifying different angle pairs, students complete a two-column proof of the Alternate Exterior Angles Theorem.



3.3.2 Guided Instruction: Angle Relationships

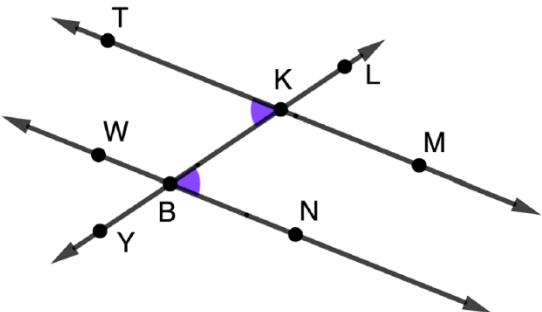
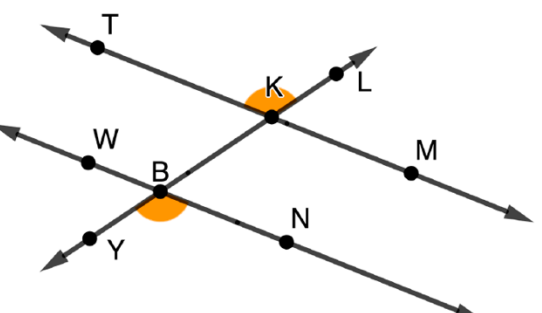
1. In each diagram, $\overleftrightarrow{TM} \parallel \overleftrightarrow{NW}$ where $\overleftrightarrow{YL}$ is a transversal.
 - a. Use patty paper to determine if the angles marked using color are congruent.

Diagram	Congruent?
A	☐ Yes ☐ No
B	☐ Yes ☐ No



For students who struggle with tracing images, consider providing copies of the two differently sized angles shown in the images for students to use in the investigation.

3.3.2 Guided Instruction: Angle Relationships (continued)

C 	☐ Yes ☐ No
D 	☐ Yes ☐ No



Before moving to the following theorems, engage students in a discussion about the placement of these four angle pairs in relation to the parallel lines and the transversal.

- Are the angles on the interior or the exterior of the parallel lines?
- Are the angles on the same side of the transversal or are they on opposite sides of the transversal (alternate)?

Having students make observations about the locations of the angles in relation to each other will help students understand the names of the angle pairs.



Which pair of angles from question 1a shows corresponding angles?

3.3.2 Guided Instruction: Angle Relationships (continued)



Alternate Exterior Angles Theorem

If two parallel lines are intersected by a transversal, then alternate exterior angles are congruent.

- b. Which diagram shows a pair of angles that are alternate exterior angles?
- ☐ Diagram A
 - ☐ Diagram B
 - ☐ Diagram C
 - ☒ Diagram D



Alternate Interior Angles Theorem

If two parallel lines are intersected by a transversal, then alternate interior angles are congruent.

- c. Which diagram shows a pair of angles that are alternate interior angles?
- ☐ Diagram A
 - ☐ Diagram B
 - ☒ Diagram C
 - ☐ Diagram D



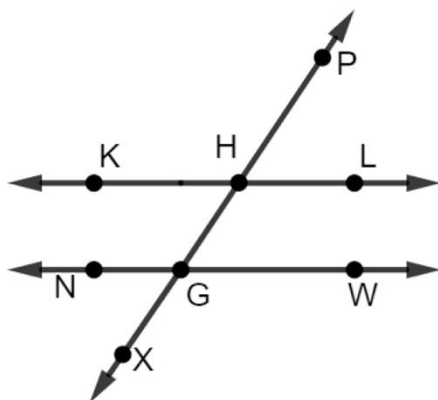
What is another pair of alternate exterior angles in diagram D?



What is another pair of alternate interior angles in diagram C?

3.3.2 Guided Instruction: Angle Relationships (continued)

2. $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$, where $\overleftrightarrow{XP}$ is a transversal.



Work with your partner to name each pair of angles as corresponding, alternate exterior, alternate interior angles, or none of these.

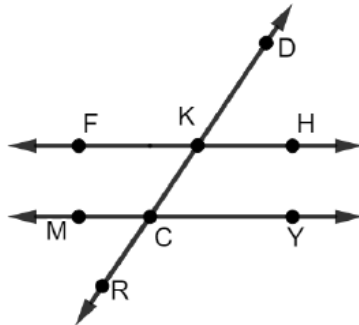
Angles	Type
$\angle WGX$ and $\angle KHP$	☐ Corresponding angles ☐ Alternate exterior angles ☐ Alternate interior angles ☐ None of the above
$\angle LHG$ and $\angle WGX$	☐ Corresponding angles ☐ Alternate exterior angles ☐ Alternate interior angles ☐ None of the above
$\angle NGX$ and $\angle KHP$	☐ Corresponding angles ☐ Alternate exterior angles ☐ Alternate interior angles ☐ None of the above
$\angle KHG$ and $\angle WGH$	☐ Corresponding angles ☐ Alternate exterior angles ☐ Alternate interior angles ☐ None of the above



For students that are having difficulties seeing and differentiating between the angle relationships, consider providing highlighters or colored markers so students can use one color for each angle pair. This will allow students to see the locations of the angles more clearly.

3.3.2 Guided Instruction: Angle Relationships (continued)

3. Given: $\overline{FH} \parallel \overline{MY}$
 Prove: $\angle DKH \cong \angle RCM$



Fill in the table to complete the proof.

Statement	Reason
1. $\overline{FH} \parallel \overline{MY}$	1. Given
2. $\angle DKH \cong \angle FKC$	2. Vertical angles are congruent.
3. $\angle CKF \cong \angle RCM$	3. If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
4. $\angle DKH \cong \angle RCM$	4. Transitive Property of Congruence



The intention of question 3 is to guide students through the completion of the two-column proof. Consider modeling the reasoning used throughout the process as a think-aloud, allowing opportunities for students to ask questions. If you prefer to release this proof to students first, consider the differentiation ideas below.



If desired, consider providing the opportunity for students to first try it with a partner on their own before reviewing whole-group. In this case, consider providing a word bank for students with the statements and reasons.

You could create the support as you see fit. For students who need a lot of support, you can provide only the statements and reasons that will be used in the proof. Then, as they go through the proof, they can cross out the ones they used.

For students who do not need as much of the support, you can still provide statements and reasons, but also include statements and reasons that won't be used. Some examples to include that will not be used in the proof could be:

- $\angle MCR \cong \angle KCY$
- $\angle DKF \cong \angle YCR$
- Linear pairs are supplementary
- Corresponding angles are supplementary

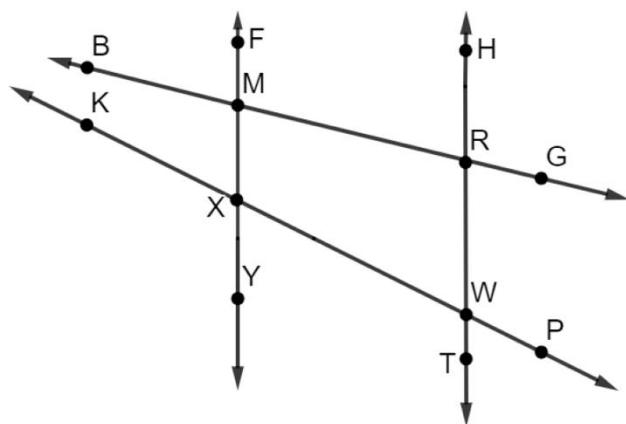
3.3.3 Try It: Angles and Parallel Lines

Component Overview: Students will work to identify the measures of angles created by the intersection of a transversal and two parallel lines and provide a justification of their reasoning.



3.3.3 Try It: Angles and Parallel Lines

1. $\overleftrightarrow{FY} \parallel \overleftrightarrow{HT}$ where $\overleftrightarrow{BG}$ and $\overleftrightarrow{KP}$ are transversals and $m\angle BMF = 76^\circ$ and $m\angle KXY = 117^\circ$.



Complete the table.

Angle	Angle Measure
$\angle MRW$	104°
$\angle PWT$	63°
$\angle YXW$	63°
$\angle RMX$	76°
$\angle PWR$	117°



Notice and Wonder

Consider implementing this strategy for students to share what they notice and wonder before completing the table. Allowing students to make sense of the image provides an entry point into the problem.

Things that students may notice include:

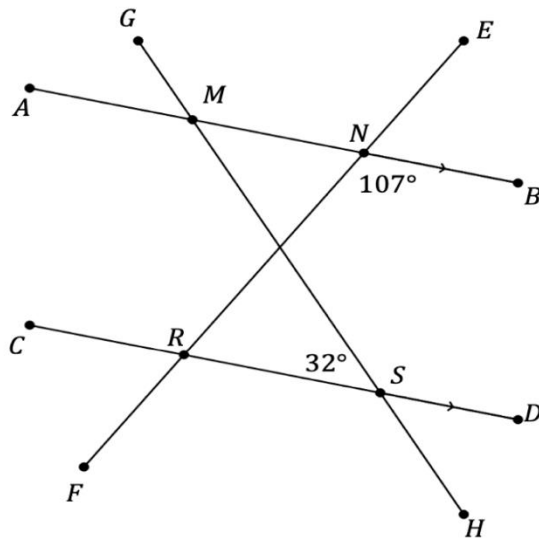
- Only two of the lines are parallel.
- There are two different transversals.
- The angle measures formed by one transversal are not congruent to the angles formed by the other transversal.
- There is a quadrilateral in the middle of the image, but it is not a parallelogram (this is an important observation that will be explored later in the course).



For students who are having difficulties finding congruent angles, have them cover up one of the transversals to find special angle pairs formed by the two parallel lines and one of the transversals.

3.3.3 Try It: Angles and Parallel Lines (continued)

2. $\overline{AB} \parallel \overline{CD}$ where $\overline{GH}$ and $\overline{EF}$ are transversals and $m\angle BNR = 107^\circ$ and $m\angle RSM = 32^\circ$.



Complete the table.

Angle Measure	Postulate or Theorem Used
$m\angle NRC = 107^\circ$	<i>Alternate Interior Angles Theorem</i>
$m\angle BMH = 32^\circ$	<i>Alternate Interior Angles Theorem</i>
$m\angle DRF = 107^\circ$	<i>Corresponding Angles Theorem or Vertical Angles Theorem</i>
$m\angle GMA = 32^\circ$	<i>Corresponding Angles Theorem or Vertical Angles Theorem</i>



For a challenge, have students label the point where $\overline{MS}$ and $\overline{NR}$ meet as point Q. Then, have them determine if it is possible to find $m\angle MQN$? Have students support their answers with angle relationships.

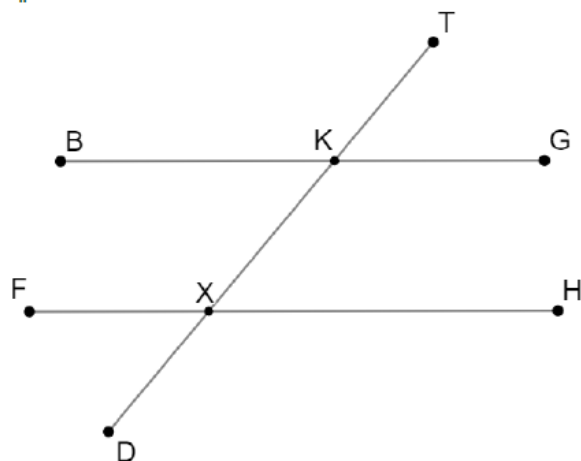
3.3.4 Your Turn!

Component Overview: Students will work to identify the angle measures and the angle relationship of angles created by the intersection of a transversal and two parallel lines.



3.3.4 Your Turn!

1. $\overline{BG} \parallel \overline{FH}$ where $\overline{TD}$ is a transversal and $m\angle HXD = 134^\circ$.



Find the measure of each angle. Then, state the angle relationships used to determine the value.

Answers will vary.

Angle Measure	Angle Relationships
$m\angle GKK = 134^\circ$	<i>Corresponding Angles</i>
$m\angle FXX = 134^\circ$	<i>Vertical Angles</i>
$m\angle BKT = 134^\circ$	<i>Alternate Exterior Angles</i>



Consider using this time to work with small groups or individuals that are still having difficulties identifying these relationships and using them to solve problems.



After we find the measure of these three angles, are we able to find the measure of the rest of the angles? Explain and support your answer.

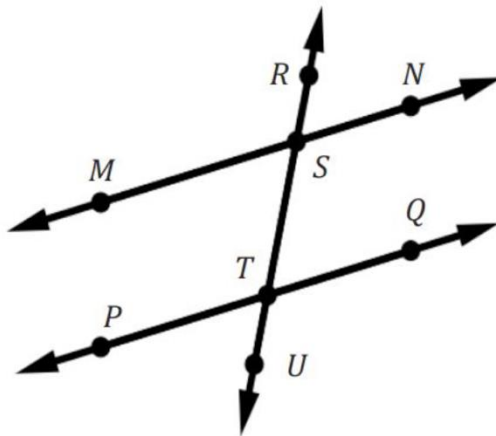
3.3.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will demonstrate their understanding of angle pair relationships created by two parallel lines intersected by a transversal.



3.3.5 Wrap-Up: Ticket Out the Door

1. $\overleftrightarrow{MN} \parallel \overleftrightarrow{PQ}$ and $m\angle PTS = 100^\circ$.



- In relation to the parallel lines, what angle relationship do $\angle TSN$ and $\angle PTS$ have?
They are alternate interior angles. They are congruent angles.
- What is $m\angle TSN$?
100°



Proficiency on this Wrap-Up includes:

- Identifying the angle relationship between two angles
- Using the angle relationship to determine the measure of an unknown angle



Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.